

Solution to Exercise 10 — Analysis on Manifolds, 80416

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Question 1

From Stokes' Theorem

$$\int_C F \, d\ell = \int_{\Sigma} \langle \text{Curl } F, N \rangle \, d\sigma$$

Subquestion 1

We consider the vector field $F(x, y, z) = (y, x - z, x^2 e^z + y^2 e^z)$ and $C = \{(x, y, z) \in \mathbb{R}^3 \mid y = 0, x^2 + z^2 = 1\}$.

We will calculate $\int_C F \, d\ell$ where C is oriented counterclockwise when viewed from the positive direction of the y -axis.

Solution: In our case, C is the unit circle in the xz -plane so it bounds the flat unit disk $\Sigma = \{(x, 0, z) \in \mathbb{R}^3 \mid x^2 + z^2 \leq 1\}$.

Since we look at C from the positive y -axis, by the right hand rule $N = (0, 1, 0)$ and from definition

$$\begin{aligned} \text{Curl } F &= (2ye^z + 1, -2xe^z, 0) \\ \langle \text{Curl } F, N \rangle &= -2xe^z \end{aligned}$$

Finally we need to compute the double integral

$$\iint_{\Sigma} -2xe^z \, dx \, dz$$

We notice that Σ is symmetric across the z -axis and $-2xe^z$ is an odd function with respect to x and we know that the integral of an odd function on symmetric domain is exactly 0 hence

$$\int_C F \, d\ell = 0$$

□

Subquestion 2

Let $a, b, c, d \in \mathbb{R}$ and assume that $c \neq 0$. Let γ be a circle of radius 1 contained in the plane $ax + by + cz = d$ oriented counterclockwise when viewed from the positive direction of the z -axis and we define $F(x, y, z) = (z, x, y)$. We will compute $\int_{\gamma} F \, d\ell$

Solution: As done in the lectures/restrictions, the normal vector to the given plane is the unit normal $N = \pm \frac{(a, b, c)}{\sqrt{a^2 + b^2 + c^2}}$ and since the curve is oriented counterclockwise when viewed from the positive z -axis, the z -component of the normal must be strictly positive which means that $N = \frac{\text{sgn}(c)(a, b, c)}{\sqrt{a^2 + b^2 + c^2}}$ and $\text{Curl } F = (1, 1, 1)$ so since $c \neq 0$ and the assumption of oriented counterclockwise etc

$$\langle \text{Curl } F, N \rangle = \frac{\text{sgn}(c)(a + b + c)}{\sqrt{a^2 + b^2 + c^2}}$$

That means that our integral is just a constant hence

$$\iint_{\Sigma} \langle \text{Curl } F, N \rangle \, d\sigma = \frac{\text{sgn}(c)(a + b + c)}{\sqrt{a^2 + b^2 + c^2}} \iint_{\Sigma} 1 \, d\sigma$$

From the assumption, our surface is a flat circle of radius 1. The area of a circle with $r = 1$ is exactly $\pi(1)^2 = \pi$ hence

$$\int_C F \, d\ell = \frac{\text{sgn}(c)(a + b + c)}{\sqrt{a^2 + b^2 + c^2}} \iint_{\Sigma} 1 \, d\sigma = \pi \frac{\text{sgn}(c)(a + b + c)}{\sqrt{a^2 + b^2 + c^2}}$$

□

Question 2

Let $U \subseteq \mathbb{R}^3$ be an open set and let $F : U \rightarrow \mathbb{R}^3$ be a vector field. For $p \in U$ we will show that

$$\text{Curl } F = \lim_{r \rightarrow 0} \frac{1}{|B_r(p)|} \int_{\partial B_r(p)} (N \times F) \, d\sigma$$

Where N is the outward pointing unit normal of $B_r(p)$.

Proof: Let $c = (c_1, c_2, c_3) \in \mathbb{R}^3$ from the integral linearity we have

$$c \cdot \int_{\partial B_r(p)} (N \times F) \, d\sigma = \int_{\partial B_r(p)} c \cdot (N \times F) \, d\sigma$$

We have the scalar triple product identity to get that $c \cdot (N \times F) = (F \times c) \cdot N$ so we have the vector field $G = F \times c$ and now we can use the Divergence Theorem

$$\int_{\partial B_r(p)} c \cdot (N \times F) \, d\sigma = \int_{\partial B_r(p)} (F \times c) \cdot N \, d\sigma = \int_{B_r(p)} \text{div}(F \times c) \, dx$$

We compute $\text{div}(F \times c)$, we use the identity

$$\text{div}(A \times B) = B \cdot \text{div}(A) - A \cdot \text{div}(B)$$

We can prove it but I think Dan showed that in the lecture anyway this is pure algebraic and from definitions.

In our case, $B = c$ so $\text{div}(B) = 0$ and we have

$$\text{div}(F \times c) = c \cdot \text{div}(F) = c \cdot \text{Curl } F$$

Finally we have

$$\int_{\partial B_r(p)} c \cdot (N \times F) \, d\sigma = \int_{B_r(p)} c \cdot \text{Curl } F \, dx = c \cdot \int_{B_r(p)} \text{Curl } F \, dx$$

So we got

$$c \cdot \int_{\partial B_r(p)} (N \times F) \, d\sigma = c \cdot \int_{B_r(p)} \text{Curl } F \, dx$$

But this is true for every vector which means that

$$\int_{\partial B_r(p)} (N \times F) \, d\sigma = \int_{B_r(p)} \text{Curl } F \, dx$$

Which actually means since $\text{Curl } F$ is continuous

$$\lim_{r \rightarrow 0} \frac{1}{|B_r(p)|} \int_{\partial B_r(p)} (N \times F) \, d\sigma = \lim_{r \rightarrow 0} \frac{1}{|B_r(p)|} \int_{B_r(p)} \text{Curl } F \, dx \stackrel{\text{Mean Value Theorem}}{=} \lim_{x_r \rightarrow p} \text{Curl } F(x_r) = \text{Curl } F(p)$$

□

Question 3

Let $0 < k < d$ and we assume that $M \subseteq \mathbb{R}^d$ is a k -dimensional manifold with boundary. We will prove that ∂M is $(k - 1)$ -dimensional manifold without boundary.

Proof: Since M is a k -dimensional manifold for every $p \in M$ there exists an open neighborhood $p \in U \subseteq \mathbb{R}^d$ and a diffeomorphism $\varphi : U \cap M \rightarrow V \subseteq H^k := \{(x_1, \dots, x_k) \in \mathbb{R}^k \mid x_k \geq 0\}$.

Assume that $p \in \partial M$ by definition of the boundary of a manifold with boundary, we may assume (after possibly shrinking U) that

$$\varphi((U \cap M) \cap \partial M) = V \cap \partial H^k$$

Where

$$\partial H^k = \{(x_1, \dots, x_k) \in \mathbb{R}^k \mid x_k = 0\}$$

We consider $\psi := \varphi|_{U \cap \partial M} : U \cap \partial M \rightarrow V \cap \partial H^k$ and since φ is a diffeomorphism, ψ is a homeomorphism onto its image. Since $V \cap \partial H^k$ is open in ∂H^k , there exists an open set $W \subseteq \mathbb{R}^k$ such that $V = W \cap H^k$ therefore

$$V \cap \partial H^k = (W \cap H^k) \cap \partial H^k = W \cap \partial H^k$$

Which is open in ∂H^k and we define the map $\pi : \partial H^k \rightarrow \mathbb{R}^{k-1}$ by $\pi(x_1, \dots, x_{k-1}, 0) = (x_1, \dots, x_{k-1})$ which is diffeomorphism, hence $\pi \circ \psi : U \cap \partial M \rightarrow \pi(V \cap \partial H^k)$ is a homeomorphism onto an open subset of $\mathbb{R}^{k-1}$.

Thus every point $p \in \partial M$ possesses a neighborhood in ∂M that is diffeomorphic to an open subset of $\mathbb{R}^{k-1}$ which means that ∂M is a $(k - 1)$ -dimensional manifold.

Furthermore, since $V \cap \{x_k = 0\} \cong \mathbb{R}^{k-1}$ the manifold is without a boundary. □

Question 4

Let $S \subseteq \mathbb{R}^3$ be a surface without a boundary and let $p_0 \in \mathbb{R}^3$.

We consider the function $f : S \rightarrow \mathbb{R}$ defined by $f(p) = \|p - p_0\|^2$ and we will prove that f is smooth and for every $w \in T_p S$ we have $Df_p(w) = \langle 2w, p - p_0 \rangle$.

Proof: For the smoothness part, if we look at $\tilde{f} : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ given by $\tilde{f}(x) = \langle x - p_0, x - p_0 \rangle$ we get that $\tilde{f}$ is a polynomial therefore smooth and $\tilde{f}|_S = f$ therefore smooth.

Let $w \in T_p S$, by definition there exists a smooth curve $\gamma : (-\varepsilon, \varepsilon) \rightarrow S$ such that $\gamma(0) = p$ and $\dot{\gamma}(0) = w$ so we compute

$$Df_p(w) = \frac{d}{dt} f(\gamma(t))|_{t=0} = \frac{d}{dt} \langle \gamma(t) - p_0, \gamma(t) - p_0 \rangle |_{t=0} = 2 \langle \dot{\gamma}(0), \gamma(0) - p_0 \rangle = \langle 2w, p - p_0 \rangle$$

□

Question 5

Let $L : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be a linear map and let $S \subseteq \mathbb{R}^3$ be a surface without a boundary invariant under L (i.e. $L(S) \subseteq S$). We will prove that $L|_S : S \rightarrow S$ is smooth and that $\forall p \in S, \forall w \in T_p S, DL_p(w) = L(w)$.

Proof: For the smooth part is driven from the fact that L is linear which means its components are degree-1 polynomials hence smooth therefore the restriction is smooth.

Let $w \in T_p S$ and let γ be a curve in S with $\gamma(0) = p, \dot{\gamma}(0) = w$.

By definition, the differential $DL_p(w)$ is simply the velocity of this new curve at time $t = 0$ so we have

$$DL_p(w) = \frac{d}{dt} L(\gamma(t))|_{t=0} \stackrel{L \text{ is linear}}{=} L\left(\frac{d}{dt} \gamma(t)|_{t=0}\right) = L(\dot{\gamma}(0)) = L(w)$$

□

Question 6

Let $S^2 \subseteq \mathbb{R}^3$ and denote by $\psi : (0, \pi) \times (0, 2\pi) \rightarrow S^2$ the spherical coordinates $\psi(\theta, \phi) = (\sin(\theta) \cos(\phi), \sin(\theta) \sin(\phi), \cos(\theta))$. Let $\gamma : [0, 1] \rightarrow S^2$ be a smooth path $\gamma(t) = \psi(\theta(t), \phi(t))$.

We will show that

$$l(\gamma) = \int_0^1 \left(\dot{\theta}^2(t) + \dot{\phi}^2(t) \sin^2(\theta(t)) \right)^{\frac{1}{2}} dt$$

Solution: We know that

$$l(\gamma) = \int_0^1 \|\dot{\gamma}(t)\| dt$$

So using the chain rule we have

$$\dot{\gamma}(t) = \frac{\partial \psi}{\partial \theta} \dot{\theta} + \frac{\partial \psi}{\partial \phi} \dot{\phi}$$

We have

$$\begin{aligned} \frac{\partial \psi}{\partial \theta} &= (\cos(\theta) \cos(\phi), \cos(\theta) \sin(\phi), -\sin(\theta)) \\ \frac{\partial \psi}{\partial \phi} &= (-\sin(\theta) \sin(\phi), \sin(\theta) \cos(\phi), 0) \end{aligned}$$

We notice that

$$\left\langle \frac{\partial \psi}{\partial \theta}, \frac{\partial \psi}{\partial \phi} \right\rangle = -\cos(\theta) \cos(\phi) \sin(\theta) \sin(\phi) + \cos(\theta) \sin(\phi) \sin(\theta) \cos(\phi) + 0 = 0$$

Which mean that if we set $e_1 = \frac{\partial \psi}{\partial \theta}$, $e_2 = \frac{\partial \psi}{\partial \phi}$ they form an orthogonal basis for the tangent space so we compute

$$\begin{aligned} \|e_1\|^2 &= \cos^2(\theta) \cos^2(\phi) + \cos^2(\theta) \sin^2(\phi) + \sin^2(\theta) = \cos^2(\theta) \underbrace{(\cos^2(\phi) + \sin^2(\phi))}_{=1} + \sin^2(\theta) = \cos^2(\theta) + \sin^2(\theta) = 1 \\ \|e_2\|^2 &= \sin^2(\theta) \sin^2(\phi) + \sin^2(\theta) \cos^2(\phi) = \sin^2(\theta) \underbrace{(\sin^2(\phi) + \cos^2(\phi))}_{=1} = \sin^2(\theta) \end{aligned}$$

We remember that $\|v\|^2 = \langle v, v \rangle$ so in our case $v = \dot{\theta}e_1 + \dot{\phi}e_2$ so from identity in Functional calculate we will have

$$\|\dot{\theta}e_1 + \dot{\phi}e_2\|^2 = \dot{\theta}^2 \langle e_1, e_1 \rangle + 2\dot{\theta}\dot{\phi} \langle e_1, e_2 \rangle + \dot{\phi}^2 \langle e_2, e_2 \rangle \underset{\langle e_1, e_1 \rangle = \|e_1\|^2}{=} \dot{\theta}^2 \|e_1\|^2 + 2\dot{\theta}\dot{\phi} \langle e_1, e_2 \rangle + \dot{\phi}^2 \|e_2\|^2 \underset{\langle e_1, e_2 \rangle = 0}{=} \|e_1\|^2 \dot{\theta}^2 + \|e_2\|^2 \dot{\phi}^2$$

So from the above we get that

$$\|\dot{\gamma}(t)\|^2 = \dot{\theta}^2 + \dot{\phi}^2 \sin^2(\theta)$$

Since $\gamma : [0, 1] \rightarrow S^2$ all that is left is taking the square root to achieve the wanted result. □